

# SOOJIN HWANG

AI RESEARCHER / ENGINEER



## CONTACT

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- developing-soosoo

## EDUCATION

2026.02 - 2024.02  
POSTECH

- MSc. in Artificial Intelligence
- GPA: 3.93/4.30

2019.03 - 2024.02  
INHA UNIVERSITY

- BSc. in Computer Science and Engineering
- GPA: 4.22/4.50

2022.09 - 2023.01  
UNIVERSITY OF HULL

- Computer Science
- Artificial Intelligence

## SKILLS

- Languages
  - Python, C++, SQL, JAVA
- ML/DL
  - PyTorch, TensorFlow, Transformers, Hugging Face
- Tools
  - Git, AWS, Linux, Notion

## LANGUAGES

- English (Fluent)
- Japanese (Basics)

## PROFILE

I received my Master's degree in Artificial Intelligence at POSTECH under the supervision of Professor Won Hwa Kim. My research focuses on multi-modal learning and computer vision, leveraging large-scale real-world data to build practical AI systems, including applications in medical and semiconductor domains.

## PUBLICATIONS

- Soojin Hwang\***, Jaeyoon Sim\*, Won Hwa Kim, "HiMix: Hierarchical Visual-Textual Mixing Network for Lesion Segmentation", IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026 (\*Equal contribution)
- Soojin Hwang\***, Jaeyoon Sim\*, Won Hwa Kim, "REMix: Refinement-Enhanced Visual-Textual Mixing Network for Lesion Segmentation", International Workshop on Emerging LLM/LMM Applications in Medical Imaging (ELAMI), 2026 (\* Equal contribution)
- Jaeyoon Sim, **Soojin Hwang**, Seunghun Baek, Guorong Wu, Won Hwa Kim, "Learning Multi-Scale Hypergraph for High-Order Brain Connectivity Analysis", International Conference on Machine Learning (ICML), 2026.
- Minjae Lee\*, Sungwoo Hur\*, **Soojin Hwang**, Won Hwa Kim, "PR-MaGIC: Prompt Refinement Via Mask Decoder Gradient Flow For In-Context Segmentation", Computer Vision and Pattern Recognition (CVPR), 2026
- Yechan Hwang, **Soojin Hwang**, Guorong Wu, Won Hwa Kim, "Multi-order simplex-based graph neural network for brain network analysis", MICCAI (International Conference on Medical Image Computing and Computer-Assisted Intervention), 2024
- Soojin Hwang**, Donghyun Kim, Young-Duk Seo, "A BERT-Based Masked Language Model Using a Morphological Analyzer", Proceedings of KIIT Conference, 2023
- Jaeyoon Sim\*, **Soojin Hwang\***, Guorong Wu, Won Hwa Kim, "Transformer-Guided Adaptive Diffusion for Multi-Modal Brain Network Analysis", Submitted to IEEE Transactions on Medical Imaging (Under Review) (\*Equal contribution)

# AWARDS & HONORS

## Excellent Paper Award, 39th KSIIM Annual Conference (Jul 2025)

Recognized with the Excellence Award at the 39th KSIIM Annual Conference

## Grand Prize for Capstone Design, Inha University (Jun 2023)

Awarded the Highest Excellence Prize for developing an "AI-Powered Presentation Analysis Service"

## Best Paper Award, KIIT Research Paper Competition (Jun 2023)

Received the Outstanding Paper Award with the paper "A BERT-Based Masked Language Model Using a Morphological Analyzer"

# WORK EXPERIENCE

## Video Streaming & Personal Protective Equipment (PPE) Detection

Korea Institute of Science and Technology (KIST), Winter Internship

Seoul, Korea  
Dec 2021 - Feb 2022

- Build a cloud based AI video analysis pipeline with AWS services (Kinesis Video Stream, Rekognition, S3)
- Enabled real-time video streaming from a Double 3 robot camera using GStreamer on Ubuntu.

# RESEARCH PROJECTS

## Semiconductor Layout GDS(Graph Design System) Understanding Model

POSTECH, joint research with Samsung Electronics DS

Pohang, Korea  
Apr 2025 - Feb 2026

- Exploring two approaches (1) LLM-based and (2) Point-based multimodal, developed a unified semiconductor layout understanding framework to address various GDS-based tasks.
- Designed domain-specific pretext tasks for pre-training and established evaluation benchmarks to validate geometric understanding.

## Prediction of Alzheimer's Disease Progression with Diagnostic Reports

POSTECH, Creative Research Project of ITRC-hCIA

Pohang, Korea  
Sep 2024 - Mar 2025

- Redefined Clinical Dementia Rating (CDR) prediction as an image-language matching problem using brain MRI.
- Implemented an ordinal regression framework based on OrdinalCLIP to preserve dementia severity rank attributes.

## Delving into the Depths

POSTECH, Term Project of Course Work, Deep Learning

Pohang, Korea  
Feb 2024 - Jun 2024

- Report "Delving into the Depths : Investigating Training Convergence Across Varying Network Depths."
- Explored the conterintuitive phenomenon where deeper neural networks converge faster than shallow ones.
- Tested two hypothesis involving gradient sharpness and initialization distance

## SNS Comment based Sentiment Analysis Model

University of Hull, Exchange Student Coursework, Aritifial Intelligence

Hull, England  
Sep 2022 - Jan 2023

- Constructed a GRU-based model using TensorFlow/Keras, implementing domain-specific text preprocessing (URL/handle removal, tokenization, padding).

# CERTIFICATES

|      |                                                       |                                              |
|------|-------------------------------------------------------|----------------------------------------------|
| 2026 | TOEIC Speaking Advanced Low (AL)                      | ETS / YBM                                    |
| 2026 | Engineer Information Processing (written examination) | Human Resources Development Service of Korea |
| 2026 | Advance Data Analytics Semi-Professional (ADsP)       | Korea Data Industry Promotion Agency         |
| 2026 | SQL Developer (SQLD)                                  | Korea Data Industry Promotion Agency         |

# TEACHING EXPERIENCES

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## Teaching Assistant for Computer Vision

POSCO Youth AI Big Data Academy

Pohang, Korea  
Dec 2024 - Dec 2024

- Assisted in teaching computer vision topics, focusing on Generative Models (GANs and Diffusion Models).
- Managed PyTorch and Jupyter Notebook environment setups and provided technical troubleshooting.

## Teaching Assistant

POSTECH

Pohang, Korea  
Sep 2025 - Dec 2025

- AIGS538 : Deep Learning (Fall 2025)

# RESEARCH & ACADEMIC EXPERIENCE

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## Undergraduate Research Assistant

KDD (Knowledge Discovery and Data Mining) Lab, Inha University

Incheon, Korea  
Feb 2023 - Jan 2024

- Conducted foundational research in Natural Language Processing (NLP) with a focus on specialized Korean language processing models.
- Led a NLP study group based on the Stanford CS224N: Natural Language Processing with Deep Learning.

## LikeLion Club (University IT & Software Development)

Member

Incheon, Korea  
Mar 2020 - Dec 2020

- Developed React frontend applications and implemented Python data structures/algorithms
- Engineered an beta-testing feedback platform during the Hackathon to facilitate agile product iterations.

## Academic Engineering Clubs

Team Leader & Member

Incheon, Korea  
Jun 2020 - Jun 2022

- Team Leader | **WEBS (Programming Club)** : Spearheaded a C++ programming study group, designing an curriculum and mentoring team members to strengthen their engineering and programming capabilities.
- Member | **IGRUS (Computer & Security Club)** : Developed a simulation game using C# and Unity; participated in cybersecurity studies covering security fundamentals and cryptographic principles.

# EXTRACURRICULAR ACTIVITIES

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## Personal Engineering Blog (Tistory)

Blogger

Korea  
Aug 2021 - Present

- Maintained a personal engineering blog for 5 years, archiving research paper reviews, algorithm analyses, and structured technical content to internalize core CS/AI concepts.

## The 23rd Moonlight Night of Silla 165-Ri Walking Competition

Finisher / Participant

Gyeongju, Korea  
Jun 14, 2025 - Jun 15, 2025

- Successfully completed the designated 75-Ri (30km) long-distance course

## MIP Lab Futsal Group & Pingu (Women's Futsal Club)

Team Member

Pohang, Korea  
Mar 2024 - Feb 2026

- Promoted a healthy, collaborative research environment and enhanced teamwork